

L12: Regression Techniques Notes

Unbiasedness

The linear model is:

$$\underset{\sim}{y} = X \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}, \quad E(\underset{\sim}{\varepsilon}) = \underset{\sim}{0}, \quad \text{Var}(\underset{\sim}{\varepsilon}) = \sigma^2 I$$

$$\underset{\sim}{\varepsilon} \sim N(0, \sigma^2 I)$$

$$E(\underset{\sim}{\varepsilon}) = \underset{\sim}{0} \quad \times$$

Consider predictors $\{x_1, \dots, x_p\}$ and *missing* variables $\{z_1, \dots, z_q\}$, with the $n \times q$ matrix:

$$\underset{n \times q}{Z} = \begin{bmatrix} z_{\sim 1} & \cdots & z_{\sim q} \end{bmatrix}$$

Two Working Models

$$(A) \quad \underset{\sim}{y} = X \underset{\sim}{\beta} + Z \underset{\sim}{\gamma} + \underset{\sim}{\varepsilon}^*, \quad \underset{\sim}{\varepsilon}^* \sim N(0, \sigma^2 I)$$

$$W = [X : Z] \quad (n \times p+1)$$

$$(b) \quad \underset{\sim}{y} = X \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}, \quad \underset{\sim}{\varepsilon} \sim N(0, \sigma^2 I)$$

We have observed that:

$\underset{\sim \text{OLS}}{\hat{\beta}}$ is **biased**, but has *same variance* as before.

OLS Estimator Under Models (A) and (b)

$$\begin{aligned} \underset{\sim \text{OLS}}{\hat{\beta}} &= (X'X)^{-1} X' \underset{\sim}{y} \\ &\stackrel{(b)}{=} \underset{\sim}{\beta} + (X'X)^{-1} X' \underset{\sim}{\varepsilon} \\ &\stackrel{(A)}{=} \underset{\sim}{\beta} + \underbrace{(X'X)^{-1} X' Z \underset{\sim}{\gamma}}_{\text{location shift}} + (X'X)^{-1} X' \underset{\sim}{\varepsilon}^* \end{aligned}$$

Similarly, $\underset{\sim \text{OLS}}{\hat{y}}$ is a **biased** estimate of $X \underset{\sim}{\beta}$, but has the same variance.

$$E\left(\frac{RSS}{n-p-1}\right) = E(S_n) = \sigma^2 + \frac{\gamma^T Z^T (I - P(X)) Z \gamma}{n-p-1} \geq \sigma^2$$

RSS Expansion

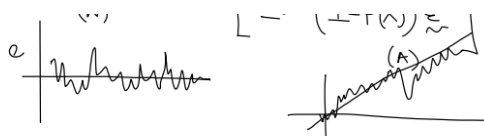
$$RSS = \underset{\sim}{y}^T (I - P(X)) \underset{\sim}{y}$$

$$\stackrel{(b)}{=} \underset{\sim}{\varepsilon}^T (I - P(X)) \underset{\sim}{\varepsilon}$$

$$\stackrel{(A)}{=} \underset{\sim}{\varepsilon}^{*T} (I - P(X)) \underset{\sim}{\varepsilon}^* + \underset{\sim}{\gamma}^T Z^T (I - P(X)) Z \underset{\sim}{\gamma} + 2 \underset{\sim}{\gamma}^T Z^T (I - P(X)) \underset{\sim}{\varepsilon}^*$$

$$\underset{\sim}{e} = (I - P(X)) \underset{\sim}{y} \stackrel{(A)}{=} (I - P(X)) Z \underset{\sim}{\gamma} + (I - P(X)) \underset{\sim}{\varepsilon}^*$$

$$\stackrel{(b)}{=} \left[(I - P(X)) \underset{\sim}{\varepsilon} \right]$$



Fitted Values

New Observation

New observation: $(\underset{\sim}{x}', \underset{\sim}{z}')$ where $\underset{\sim}{z}$ is not observed.

$$y_0 = \underset{\sim}{x}_0^T \underset{\sim}{\beta} + \underset{\sim}{z}_0^T \underset{\sim}{\gamma} + \underset{\sim}{\varepsilon}_0^* \quad \text{--- (A)}$$

$$\hat{y}_{0,\text{ols}} = \underset{\sim}{x}_0^T \underset{\sim}{\hat{\beta}}_{\text{ols}}$$

$$E(\hat{y}_{0,\text{ols}}) = \underset{\sim}{x}_0^T \underset{\sim}{\beta} \neq E(y_0) = \underset{\sim}{x}_0^T \underset{\sim}{\beta} + \underset{\sim}{z}_0^T \underset{\sim}{\gamma}$$

$$\text{Var}(\hat{y}_{0,\text{ols}}) = \underset{\sim}{x}_0^T (X'X)^{-1} \underset{\sim}{x}_0 \sigma^2$$

If Model (A) Were Chosen as Working Model

$$\hat{y}_{0,\text{ols}}^{(A)} = \begin{pmatrix} x'_{\sim 0} & z'_{\sim 0} \end{pmatrix} \hat{\delta}_{\sim \text{ols}}, \quad \hat{\delta}_{\sim \text{ols}} = (W^T W)^{-1} W^T y_{\sim}$$

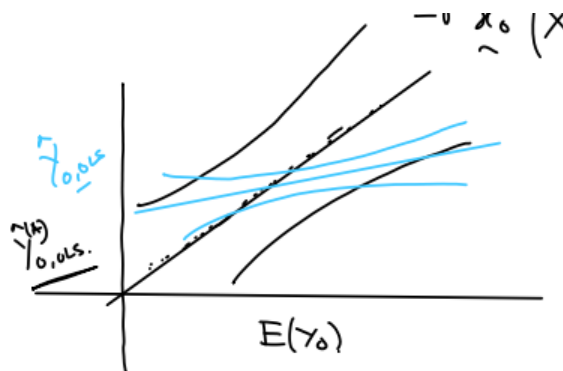
$$E\left(\hat{y}_{0,\text{ols}}^{(A)}\right) = \begin{pmatrix} x_{\sim 0}^T & z_{\sim 0}^T \end{pmatrix} \delta_{\sim} = E(y_0)$$

where $W = [X : Z]$ and $\delta_{\sim} = \begin{bmatrix} \beta \\ \gamma \end{bmatrix}$.

Variance of Fitted Values Under Model (A)

$$\begin{aligned} \text{Var}(\hat{y}_{0,\text{ols}}) &= \begin{pmatrix} x'_{\sim 0} & z'_{\sim 0} \end{pmatrix} (W^T W)^{-1} \begin{pmatrix} x_{\sim 0} \\ z_{\sim 0} \end{pmatrix} \\ &= \sigma^2 x_{\sim 0}^T (X'X)^{-1} x_{\sim 0} + \sigma^2 \frac{(a_{\sim 0} - z_{\sim 0})^T H^{-1} (a_{\sim 0} - z_{\sim 0})}{\sim 0} \end{aligned}$$

where $a_0 = Z'X(X'X)^{-1}x_{\sim 0}$ and $H = Z'(I - P(X))Z$.



Including Extra Variables: What If $\{x_1, \dots, x_p\}$ Are Actual Predictors?

What if $\{x_1, \dots, x_p\}$ are the *actual* predictors, but we also include $\{z_1, \dots, z_q\}$ in the model?

$$y_{\sim} = X_{\sim} \beta_{\sim} + \varepsilon_{\sim}^*; \quad \varepsilon_{\sim}^* \sim N(0, \sigma^2 I) \quad (A^*)$$

$$y_{\sim} = X_{\sim} \beta_{\sim} + Z_{\sim} \gamma_{\sim} + \varepsilon_{\sim}; \quad \varepsilon_{\sim} \sim N(0, \sigma^2 I) \quad (W^*)$$

$$\delta_{\sim} = \begin{bmatrix} \beta_{\sim} \\ \gamma_{\sim} \end{bmatrix} \quad \hat{\delta}_{\sim \text{ols}} = (W^T W)^{-1} W^T y_{\sim} \quad \text{where } W = [X : Z]$$

Q: Is $\hat{\beta}_{\sim\text{OLS}}$ Unbiased?

$$\begin{aligned}\hat{\beta}_{\sim\text{ols}} &= [I_{p+1} \ : \ O] \hat{\delta}_{\sim\text{ols}} \\ E(\hat{\beta}_{\sim\text{ols}}) &= E_{p+1}(W^T W)^{-1} W^T \underbrace{E(y)}_{X\beta = W \begin{pmatrix} \beta \\ 0 \end{pmatrix}} \\ &= E_{p+1} \begin{pmatrix} \beta \\ 0 \end{pmatrix} = \beta\end{aligned}$$

Note:

$$\begin{aligned}\begin{bmatrix} X\beta \\ +Z0 \end{bmatrix} &= W \begin{pmatrix} \beta \\ 0 \end{pmatrix} \\ \text{Var}(\hat{\beta}_{\sim\text{ols}}) &= \text{Var}\left(E_{p+1} \hat{\delta}_{\sim\text{ols}}\right) = E_{p+1} \text{Var}(\hat{\delta}_{\sim\text{ols}}) E_{p+1}^T\end{aligned}$$

Variance Inflation from Over-Specification

$$\begin{aligned}&= \sigma^2 [I_{p+1} \ : \ O] (W^T W)^{-1} \begin{bmatrix} I_{p+1} \\ O \end{bmatrix} \\ &= \sigma^2 [\text{principal submatrix of } (W^T W)^{-1} \text{ of order } p+1]\end{aligned}$$

Comparison with $\sigma^2(X'X)^{-1}$

$\sigma^2(X'X)^{-1}$ Should be the variance of $\hat{\beta}_{\sim\text{ols}}$ if (A^*) is chosen as the working model.

$$\begin{aligned}&= \sigma^2 \left[(X'X)^{-1} + (X'X)^{-1} X^T Z \tilde{H}^{-1} Z' X (X'X)^{-1} \right] \\ &\geq \sigma^2 (X'X)^{-1}\end{aligned}$$

where $W = [X \ : \ Z]$ ($n \times p+1$) and:

$$W^T W = \begin{bmatrix} X^T X & X^T Z \\ Z^T X & Z^T Z \end{bmatrix}^{p+1 \times p+1}$$

Comparison of Square Matrices

Two square matrices A, B can be compared as follows:

We say $A \geq B$ iff (1) $A - B$ is p.s.d.

$\Leftrightarrow$ (2) All eigenvalues of $A - B$ are non-negative.

$\Leftrightarrow$ (3) All quadratic forms involving $(A - B)$ are non-negative.

Similarly, show that:

1) $\hat{y}_{\sim\text{ols}}$ has **zero-bias** but higher variance.

2) $RSS/(n - p - q - 1)$ is unbiased for σ^2 .

Heteroskedasticity

$$\text{Var}(\varepsilon_{\sim}) = \sigma^2 V$$

$$y_{\sim} = X_{\sim} \beta + \varepsilon_{\sim}$$

$$\varepsilon_{\sim} \sim N(0, \sigma^2 I) \quad \text{--- (w)}$$

$$\varepsilon_{\sim} \sim N(0, \sigma^2 V) \quad \text{--- (A)}$$

$$\hat{\beta}_{\sim \text{ols}} = (X'X)^{-1} X' y_{\sim}$$

- $E(\hat{\beta}_{\sim}) = \beta_{\sim}$
- $\text{Var}(\hat{\beta}_{\sim \text{ols}}) = \sigma^2 (X'X)^{-1} X' V X (X'X)^{-1}$

Variance Under Heteroskedasticity & RSS

$$\bullet \text{Var}(\hat{\beta}_{\sim \text{ols}}) = \sigma^2 \underbrace{(X'X)^{-1} X' V X (X'X)^{-1}}$$

Similarly, $\hat{y}_{\sim \text{as}}$ and RSS :

$$E(RSS) = E\left(y_{\sim}^T (I - P(X)) y_{\sim}\right) = E\left(\varepsilon_{\sim}^T (I - P(X)) \varepsilon_{\sim}\right) = \text{tr}\left(\text{Var}(\varepsilon_{\sim}) (I - P(X))\right) \approx \sigma^2 \text{tr}(V(I - P(X)))$$

Non-normality

$$y_{\sim} = X_{\sim} \beta + \varepsilon_{\sim}, \quad E(\varepsilon_{\sim}) = 0, \quad \text{Var}(\varepsilon_{\sim}) = \sigma^2 I$$

$$H_0 : A \beta_{\sim} = c_{\sim}$$

$$\varepsilon_{\sim} \sim N(0, \sigma^2 I) \quad \times$$

$$\text{Var}(\varepsilon_{\sim}) = \sigma^2 V$$

$$V = \begin{bmatrix} 1 & \rho & \rho^2 & \dots & \rho^{n-1} \\ \rho & 1 & \rho & \dots & \rho^{n-2} \\ \vdots & & & & \vdots \\ \rho^{n-1} & \rho^{n-2} & \rho^{n-3} & \dots & 1 \end{bmatrix} \quad \text{AR}(1), \quad |\rho| < 1$$

$$\begin{bmatrix} 1 & \rho & \cdots & \rho \\ \rho & 1 & \cdots & \rho \\ \vdots & \vdots & \ddots & \vdots \\ \rho & \rho & \cdots & 1 \end{bmatrix} \quad \text{Equi-correlation matrix}$$

$$-\frac{1}{n-1} < \rho < 1$$

Non-normality (continued) :: $\hat{\beta}_{\sim \text{OLS}}$ is BLUE

- **Testing.** (test statistics are usually based on the distribution of $\hat{\beta}_{\sim \text{OLS}}$)
- **Confidence / Prediction intervals.**

X is Non-random

When X is **random**:

$$\begin{bmatrix} y \\ x_1 \\ x_2 \\ \vdots \\ x_p \end{bmatrix} \quad \text{is a Random Variable.}$$

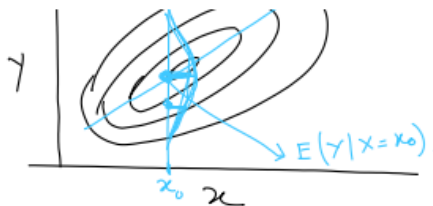
$$E(Y \mid X_1 = x_1, \dots, X_p = x_p) = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$$

$$E(Y) = \beta_0 + \beta_1 E(X_1) + \cdots + \beta_p E(X_p)$$

$$\text{Var}(Y \mid X_1 = \dots) = \sigma^2 I$$

$$Y \mid X_1 = x_1, \dots, X_p = x_p \sim N(\underset{\sim}{0}, \sigma^2 \underset{\sim}{I}) \iff Y \mid X \stackrel{D}{=} \underset{\sim}{X} \underset{\sim}{\beta} + \underset{\sim}{\varepsilon}$$

where $\underset{\sim}{\varepsilon} \sim N(\underset{\sim}{0}, \sigma^2 \underset{\sim}{I})$.



OLS When X is Random

$$\begin{aligned}
\hat{\beta}_{\sim \text{ols}} &= (X'X)^{-1}X^T y_{\sim} \\
RSS &= y_{\sim}^T (I - P(X)) y_{\sim} \\
E\left(\hat{\beta}_{\sim \text{ols}} \mid X\right) &= (X'X)^{-1}X^T \underbrace{E(y_{\sim} \mid X)}_{X\beta_{\sim}} = \beta_{\sim} \\
E(\hat{\beta}_{\sim \text{ols}}) &= E_X\left(E(\hat{\beta}_{\sim \text{ols}} \mid X)\right) = \beta_{\sim} \\
\text{Var}\left(\hat{\beta}_{\sim \text{ols}} \mid X\right) &= \sigma^2 (X'X)^{-1} \\
\text{Var}(\hat{\beta}_{\sim \text{ols}}) &= E_X\left(\text{Var}(\hat{\beta}_{\sim \text{ols}} \mid X)\right) + \text{Var}_X\left(\underbrace{E(\hat{\beta}_{\sim \text{ols}} \mid X)}_{\beta_{\sim}}\right) \\
&= \sigma^2 E((X'X)^{-1}) + 0 = \sigma^2 E((X'X)^{-1}) \\
\hat{y}_{\sim \text{ols}} &= X\hat{\beta}_{\sim \text{ols}} = P(X)y_{\sim} \\
E(\hat{y}_{\sim \text{ols}}) &= E_X\left(\underbrace{P(X)X\beta_{\sim}}_{\sim}\right) = E_X(X\beta_{\sim}) = E(X)\beta_{\sim} \\
\text{Var}(\hat{y}_{\sim \text{ols}}) &= E_X(\text{Var}(\hat{y}_{\sim \text{ols}} \mid X)) + \text{Var}(E_{Y|X}(\hat{y}_{\sim \text{ols}} \mid X)) \\
&= E_X(\sigma^2 P(X)) + \text{Var}(X\beta_{\sim}) = \sigma^2 E_X(P(X)) + \text{Var}(X\beta_{\sim})
\end{aligned}$$

$$\begin{aligned}
&E\left((X\beta_{\sim} - E(X)\beta_{\sim})(X\beta_{\sim} - E(X)\beta_{\sim})^T\right) \\
\text{Var}_{y|x}\left(\hat{y}_{\sim \text{ols}}\right) &= \text{Var}_{y|x}(P(X)y_{\sim}) = \sigma^2 P(X)
\end{aligned}$$

$$\begin{aligned}
E(RSS) &= E_X\left(E_{y|x}\left(y_{\sim}^T (I - P(X)) y_{\sim}\right)\right) = E_X\left[E_{y|x}\left(\varepsilon_{\sim}^T (I - P(X)) \varepsilon_{\sim}\right)\right] \\
&[\text{as } (I - P(X))X = 0 \quad \text{w.p.1}]
\end{aligned}$$

$$= E_X[\text{tr}((I - P(X)) \sigma^2 I)] = \sigma^2 E_X\left[\text{tr}\left(\underbrace{I - P(X)}_{n-p-1}\right)\right] = \sigma^2(n - p - 1)$$

$\frac{RSS}{n - p - 1}$ remains an unbiased estimate of σ^2 .

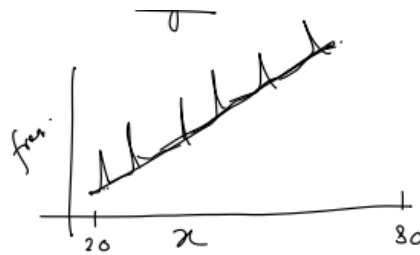
No departure from assumption.

Fixed Covariates / Predictor Observed with Error

x_j : j^{th} predictor.

We observe $x_j + \delta_j = z_j$, where δ_j is the error in observation.

Example: x : age at death.



Z : observed design matrix = $X + \Delta$
 $n \times p$ $n \times p$ $n \times p$

Actual model: $y = X\beta + \varepsilon^*$, $\varepsilon^* \sim N(0, \sigma^2 I)$.

Working model: $y = Z\beta + \varepsilon$, $\varepsilon \sim N(0, \sigma^2 I)$.

$$\hat{\beta}_{\sim \text{ols}} = (Z'Z)^{-1}Z'y_{\sim}$$

$$E(\hat{\beta}_{\sim \text{ols}}) = E_{\Delta} \left((Z'Z)^{-1}Z' \left(X\beta + \underbrace{E(\varepsilon^*)}_{(Z-\Delta)} \right) \right) = \beta_{\sim} - E_{\Delta} \left[\underbrace{(Z'Z)^{-1}Z'\Delta\beta}_{\sim} \right]$$

Bias in Errors-in-Variables Model

Need an estimate of $\Delta\beta_{\sim}$ to measure this bias.

$$\text{Var}_{y|z} \left(\hat{\beta}_{\sim \text{ols}} \right) = (Z'Z)^{-1}Z'(\sigma^2 I)Z(Z'Z)^{-1} = \sigma^2(Z'Z)^{-1}$$

$$E_{y|z}(RSS) = E \left(y_{\sim}^T (I - P(Z)) y_{\sim} \right) = E \left[(X\beta_{\sim} + \varepsilon_{\sim})^T (I - P(Z)) (X\beta_{\sim} + \varepsilon_{\sim}) \right]$$

Now, $(I - P(Z))X \neq 0$.

Collinearity

Perfect Collinearity

Perfect collinearity: $X'X$ is **singular**.

Near Collinearity

- When x_j is highly correlated with $\{x_1, \dots, x_{j-1}, x_{j+1}, \dots, x_p\}$.
- $\det(X'X)$ is very close to 0.
- $(X'X)^{-1}$ exists.

$$E\left(\hat{\beta}_{\sim \text{ols}}\right) = \beta_{\sim} \quad (\text{unbiasedness})$$

$$\text{Var}(\hat{\beta}_{\sim \text{ols}}) = \sigma^2 (X'X)^{-1}$$

$$\text{Var}(\hat{\beta}_{\sim \text{ols}}, j) = \sigma^2 e_{\sim j}^T (X'X)^{-1} e_{\sim j}$$

Assume WLOG, $j = p$

$$\begin{aligned} (X'X)^{-1} &= \begin{bmatrix} Z'Z & Z'x_{\sim p} \\ x_{\sim p}^T Z & x_{\sim p}^T x_{\sim p} \end{bmatrix}^{-1} \quad \text{where } Z = [1, \dots, x_{\sim p-1}] \\ &= \begin{bmatrix} ** & * \\ * & \left[x_{\sim p}^T (I - P(Z)) x_{\sim p} \right]^{-1} \end{bmatrix} \end{aligned}$$

$$\text{Var}(\hat{\beta}_{\sim \text{ols}, p}) = \sigma^2 e_{\sim p}^T (X'X)^{-1} e_{\sim p} = \frac{\sigma^2}{x_{\sim p}^T (I - P(Z)) x_{\sim p}}$$

If x_p is highly correlated with Z , then $(I - P(Z))x_{\sim p} = s_{\sim p}$ will have $\|s_{\sim p}\|^2$ small.

$$\|s_{\sim p}^2\|^2 = x_{\sim p}^T (I - P(Z)) x_{\sim p} \approx 0$$

Variance Inflation Factor (VIF)

Check:

$$\text{VIF}_j = \frac{\|x_{\sim j}\|^2}{x_{\sim j}^T (I - P(X_{(-j)})) x_{\sim j}} \geq 1, \quad j = 1, \dots, p$$

where $X_{(-j)} = [1_{\sim} : x_{\sim 1}, \dots, x_{\sim j-1}, x_{\sim j+1}, \dots, x_{\sim p}]$.

If VIF_j is very large for one or more j , then that indicates presence of **near collinearity**.

End of Notes